

# An Analysis on the Long Tail Impacts of Subprime Loans in the Early 2000s

Matthew Robson

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## 1 Introduction

### 1.1 Subprime Loans

Subprime Loans have been the focus of many papers on the 2008 financial crisis but are often disagreed on when it comes to specific definitions [6]. Generally, subprime loans are considered loans given to borrowers of high risk. This generally means that these loans will have higher interest rates or stricter requirements such as in the down payment or penalties. For the purposes of this paper, we will consider any loan with a rate spread greater than or equal to 3.00% as subprime. This has some drawbacks in cases where the rate spread may not indicate some other requirements of the contract or the loan or may be measured inaccurately.

Some argue that origins of subprime loans are in adjustable rate mortgages—mortgages where the interest rate for the borrower may change based on external economic circumstances [1]. Adjustable rate mortgages often lead to a predatory structure due to the increased rates over time. This coincides with subprime loans because both generally have harsher terms and so are taken out by people that have poor credit. The prevalence of subprime loans began in the early 2000s and held up to the financial crisis of 2008, after which, lending was significantly decreased. This increase in subprime loans during the early 2000s is believed to have driven an increase in construction of single family home in comparison to multi-unit housing. This is because many people that previously did not have access to taking out a loan for the construction of their own house now were given the opportunity as more loan agencies were offering subprime loans to a much wider set of people.

Subprime loans like these were then traded on the public market in many different forms such as collateralized debt obligations (CDOs) and mortgage-backed securities (MBS). A CDO is a financial device that consists of many mortgages and spreads the risk into "tranches" with differing return rates and risk rates. Generally the rating scheme for CDOs is considered highly misleading because while the senior tranches might be more secure than the mezzanine or

equity, the senior tranche is still backed by loans that may default leading to loss of profits. MBS are very similar but only directed at housing loans.

## **1.2 The 2008 Financial Crisis**

The 2008 financial crisis is often attributed to the over reliance on subprime loans and over-leveraging of the early 2000s. Because many firms in the early 2000s were buying up all of high risk subprime loans, when those borrowers eventually defaulted on their loans, the firms did not have the capital to back up their highly leveraged positions and in some cases were forced to be fully dissolved [4]. The consequences of this crisis have been seen across both the housing and economic sectors. In the housing sector, the 2008 crisis led to crash of housing prices and fire sale of a large portion of properties.

## **1.3 The Current Housing Crisis**

The current housing market is full of difficulty for new and low income buyers. Average house prices have been rising relative to average income for the past few years, leading to a high cost burden on those wanting to purchase a house. In some cases nearly 70% of a household's income is used directly on paying rent [8]. In addition to this, housing has become overcrowded such that many people are forced to live in cramped conditions [7]. There have been many proposed causes for this crisis such as the over indebtedness of people today, rapid urbanization outpacing housing development.

## **1.4 Theoretical Impacts of Subprime Loans**

# **2 Methods**

## **2.1 Data**

We collected loan data for this study from the HMDA. From our collected data, we used Loan data from the HMDA from 2004 to 2007. Data from 2000 to 2006 was sourced from OPENICPSR [3], and data from 2007 - 2010 was sourced from the Consumer Financial Protection Bureau website [2]. This data was used to generate the percent subprime loans for each census tract over this time period. Specifically, we focused on the rate spread variable to classify loans as subprime or prime. We used the ACS 5 year estimates [9] to obtain the data for our dependent variables and control variables. We used data from the Housing, Socioeconomic, and Demographic data sets for this analysis. All of the ACS data came from the 2016 - 2020 time frame. Some of the models included use spatial data for the shapes of the census tracts, and this data came from IPUMS NHGIS [5].

## 2.2 Dependent Variables

The dependent variable considered for this study was housing insecurity. While housing insecurity is a complex topic with many factors that make it up, we used cost burden and overcrowding as a proxy for housing insecurity. In this study, we consider someone to be "cost burdened" by their housing if greater than 35% of their income has gone to housing in the past 12 months. We consider the overcrowding rate of a census tract to be calculated with the following formula (Housing Units With Greater than 1 Person per Room) / (Number of Housing Units).

## 2.3 Lending Variable

The variable used for lending was rate spread. Rate spread is defined as the loan interest rate minus the average loan interest rate. We classified any value above 3% as subprime lending and any value below 3% as prime lending[2]. This variable is measured through reporting with the HMDA. When banks give out loans, they store the information about the loan and wait to report the year's loans by March 1st of the year. This results in a compiled list of every loan taken out in the year with attached location, income, demographic, and other data. From here we extracted the loans that exceed 3% in rate spread and classified those as subprime. We then found the percentage of subprime loans out of all loans taken out in a certain census tract and compiled a list of percentage subprime for each census tract.

## 2.4 Control Variables

In order to control for variability in our predicted variables, we included some control variables. We selected to include: White %, Black %, Hispanic %, Undergraduate Graduation % (of total pop.), Median Income, Unemployment Rate, and Poverty Rate. All of these variables are taken from the ACS (5yr) report from 2016 to 2020. These data comes from surveys and estimates made by ACS over this five year period [9].

# 3 Results

## 3.1 Summary Statistics

We ran our regressions on both dependent variables—overcrowding rate and cost burden. For the overcrowding rate regression, we found that the subprime loan rate was statistically significant in the baseline model, and the model including economic variables, but subprime loan rate lost significance after adding in demographic variables. On the other hand, in the cost burden regressions, we found that subprime loan rate was significant in all of our run models. Additionally, our best model achieved a  $R^2$  of  $\approx 0.404$  when predicting overcrowding rate, and  $\approx 0.298$  when predicting cost burden.

### 3.2 Regression Table 1

Table 1: Regression Results: Overcrowding Rate

|                         | <i>Dependent Variable: Overcrowding Rate</i> |                       |                        |                        |
|-------------------------|----------------------------------------------|-----------------------|------------------------|------------------------|
|                         | (1) Baseline                                 | (2) Economic          | (3) Demographic        | (4) All Variables      |
| Subprime Loan (%)       | 0.1562***<br>(0.0048)                        | 0.1029***<br>(0.0065) | 0.0062<br>(0.0074)     | 0.0024<br>(0.0073)     |
| Unemployment Rate (%)   |                                              | -0.0002**<br>(0.0001) |                        | -0.0008***<br>(0.0001) |
| Poverty Rate (%)        |                                              | 0.0009***<br>(0.0000) |                        | 0.0002***<br>(0.0000)  |
| Median Household Income |                                              | 0.0001***<br>(0.0000) |                        | 0.0001***<br>(0.0000)  |
| White (%)               |                                              |                       | -0.0013***<br>(0.0000) | -0.0013***<br>(0.0000) |
| Black (%)               |                                              |                       | -0.0012***<br>(0.0000) | -0.0011***<br>(0.0000) |
| Hispanic (%)            |                                              |                       | -0.0002***<br>(0.0000) | -0.0002***<br>(0.0000) |
| Undergraduate (%)       |                                              |                       | -0.0004***<br>(0.0000) | -0.0005***<br>(0.0000) |
| Constant                | 0.0138***<br>(0.0005)                        | 0.0039***<br>(0.0012) | 0.1438***<br>(0.0028)  | 0.1425***<br>(0.0030)  |
| Observations            | 44,611                                       | 44,489                | 44,611                 | 44,489                 |
| $R^2$                   | 0.0270                                       | 0.0581                | 0.3968                 | 0.4044                 |
| Adjusted $R^2$          | 0.0270                                       | 0.0580                | 0.3967                 | 0.4043                 |

Standard errors in parentheses.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

### 3.3 Regression Table 2

Table 2: Regression Results: Cost Burden

|                         | <i>Dependent Variable: Cost Burden Rate</i> |                         |                         |                         |
|-------------------------|---------------------------------------------|-------------------------|-------------------------|-------------------------|
|                         | (1) Baseline                                | (2) Economic            | (3) Demographic         | (4) All Variables       |
| Subprime Loan (%)       | -36.8989***<br>(1.0786)                     | -12.3028***<br>(1.4482) | -19.8165***<br>(1.7683) | -25.1691***<br>(1.6327) |
| Unemployment Rate (%)   |                                             | 0.2010***<br>(0.0158)   |                         | -0.0043<br>(0.0156)     |
| Poverty Rate (%)        |                                             | -0.0059<br>(0.0075)     |                         | -0.2112***<br>(0.0082)  |
| Median Household Income |                                             | 0.0786***<br>(0.0021)   |                         | 0.0266***<br>(0.0022)   |
| White (%)               |                                             |                         | -0.2518***<br>(0.0065)  | -0.2754***<br>(0.0066)  |
| Black (%)               |                                             |                         | -0.1558***<br>(0.0071)  | -0.1247***<br>(0.0064)  |
| Hispanic (%)            |                                             |                         | -0.0795***<br>(0.0078)  | -0.0678***<br>(0.0067)  |
| Undergraduate (%)       |                                             |                         | 0.1269***<br>(0.0036)   | 0.0525***<br>(0.0039)   |
| Constant                | 20.2390***<br>(0.1329)                      | 10.9132***<br>(0.3008)  | 32.2269***<br>(0.7134)  | 37.7072***<br>(0.6408)  |
| Observations            | 44,716                                      | 44,489                  | 44,716                  | 44,489                  |
| $R^2$                   | 0.0291                                      | 0.0722                  | 0.2389                  | 0.2981                  |
| Adjusted $R^2$          | 0.0291                                      | 0.0721                  | 0.2389                  | 0.2980                  |

Standard errors in parentheses.

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$

## 4 Discussion

The loss of predicting power and significance of subprime loan percentage when adding demographic variables to the overcrowding model is very telling. This informs us that some amount of the information encoded by race is shared by subprime percentage. This however does not include all of the information included by subprime percentage because as we see in the cost burden models, the significance of subprime percentage does not go away after adding in demographic variables. Additionally, we see the impact of subprime rate decrease after adding in the economic variables in the cost burden model. This would suggest that subprime also encodes a sort of systemic economic variable that has impacts on housing security.

The most fascinating finding here is the negative correlation between the subprime rate and cost burden. This was the opposite of what we expected but we think that this may be explained by a lack of control variables. Based on the significant reduction in correlation between subprime and cost burden after adding economic variables suggests that including more economic variables could explain other portions of that negative correlation. Specifically, considering the economic status during the loan process such as with the included income data from the HMDA data set could explain some of this negative correlation [2]. Additionally, we lack many other significant categories of control variables such as population, political, and changes in census tract shapes.

As mentioned in earlier sections, there is much more to housing insecurity than the cost burden and overcrowding that we were able to consider in this study. Variables such as homelessness, safety, and access to public facilities could all be modeled to get a more full view of the impacts of subprime loans on the current housing crisis.

## 5 Conclusion

The findings of this study show that while subprime loans may have a significant impact on housing insecurity, much more research will be needed to conclusively show the impact. Taken at face value, the study shows that census tracts with a previously greater percentage of subprime loans should expect to have lower cost burdens. However, the study also shows that census tracts with a previously greater percentage of subprime loans should expect to be more overcrowded. Together, these arguments seem to disagree, and so it is likely that more confounding variables should have been included.

This seems to implicate that investment from a bubble has both negative and positive impacts on infrastructure. Extending this to more modern examples, we might expect similar results of that of the AI bubble, and can even see parallels in dotcom bubble.

## References

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